



Rolling out mass hydroxychloroquine prophylaxis for covid-19 in India's slums risks eroding public trust

May 1, 2020

Citizens of India are in a period of a six week nationwide lockdown. Physical distancing is largely limited to the rich and upper-middle-class in urban areas and sparsely populated villages, who have the luxury of space and adequate toilet facilities. In the urban slums, however, social distancing is not possible.

The Indian Slums survey carried out in 2012, showed that India had almost 9 million households living across 33,510 urban slums. [1] The populous state of Maharashtra had almost one-quarter of these, with the state's capital Mumbai containing the largest slums in the world since the late 19th century. [2] Maharashtra makes up more than a quarter of the national aggregate of 31,332 confirmed cases of COVID-19 and more than one-third of India's recorded 1007 covid-19 related deaths as on 29th April. Recent data suggest that the major brunt of the covid-19 pandemic is being borne by those living in slum [3].

The slums of Mumbai (then known as Bombay) resulted from a complete absence of urban planning in the face of a sustained influx of migrant workers attracted to the "city of dreams" – India's financial, business and cinema capital, over the decades spanning from the end of the nineteenth century through the two world wars. Since the 1880s, the Bombay slums have been full of low-rise ill-ventilated shanty buildings. Narrow insanitary filthy lanes are occasionally flooded by the high tides from the sea flowing in through the sewage canals distributing human and animal faeces and leaving much of the city in environmental ruins during the rainy seasons [2]. Bombay has long seen unprecedented rates of mortality among the lower caste and marginalised people living in the slums. This started during the plague epidemic with 50% of increased deaths in between 1896 to 1900 and 70% in the subsequent decade. This then led to the enactment of The Epidemic Diseases Act of 1897, 50 years before India's independence. The Act, which was drawn up by the British colonial rulers, emphasized the power of the state, rather than the rights and interests of the citizens. Humanitarian provisions for security, dignity, wage compensation, and individual civil rights of quarantined individuals during pandemics, particularly poor and marginalized people from the slums, were not spelt out [4].

Dharavi in Mumbai is one of the largest and densest slums in the world. Its 1.2 million people crammed into a sprawl of barely 2.4 square kilometres during the lockdown, is about 20 times the population density of New York City. Around 400 to 500 of its residents are using each toilet complex [5]. Social distancing and isolation are therefore impossible here, as in other such locations all around the world [6] and Dharavi has seen a recent surge of COVID-19 cases with a total of 330 cases and 18 deaths as of 29th of April [3]. Within Dharavi, areas including Dr Baliga Nagar, Jasmine Mill Road, and Matunga Labor Camp, which are home to already vulnerable and marginalized communities have been marked as hotspot zones and sealed off under a municipal decree empowered by The Epidemic Diseases Act of 1897. As a

result, this is compelling people to live in abominable crowded conditions and to remain there. This is likely to increase COVID-19 transmission rates. While Dharavi is an extreme example, the most densely populated lower income group urban residential clusters in India are not dissimilar [5].

The municipal corporation of Greater Mumbai (BMC) has decided to roll out a seven-week-long course of chloroquine (CQ) and hydroxychloroquine (HCQ) mass community prophylaxis for the people living in slums [7]. The decision is apparently backed by the announcement of the Indian Council of Medical Research (ICMR) dated 22nd of March, for the prophylaxis of asymptomatic healthcare workers involved in the care of suspected or confirmed cases of covid-19 and asymptomatic household contacts of confirmed cases [8].

This is a baffling decision for while some of the studies suggest these antimalarial drugs may be effective [9–12] as well as safe [13,14], there are also concerns [15,16] that the evidence is not robust and adverse effects will be likely if the drugs are rolled out indiscriminately for mass prophylaxis, without rigorous monitoring [17]. Contradictory statements have been issued about ongoing studies and trials for CQ and HCQ prophylaxis in India [15] and concern expressed about promoting its use as a prophylactic therapy on the basis of insufficient evidence [17]. Muddled and contradictory messages about the benefits and risk of using antimalarials for mass prophylaxis to the marginalised communities in the slums are fuelling confusion and mistrust.

Absence of information regarding the risk/benefit ratio of CQ and HCQ mass prophylaxis along with proper approval from the appropriate ethical bodies needs to be shared among community health professionals and extended to the slum-dwellers community at large, instead of the ongoing uninformed prescriptive practice, which in turn puts the implementation of the proposed policy as well as other community interventions in the slum at risk. Unless convincing evidence is available, an ethically approved randomized controlled trial for studying the role of prophylactic hydroxychloroquine in preventing secondary SARS-CoV-2 infections, based on the ring vaccination design for smallpox and Ebola should be started [18].

The nationwide lockdown has already reduced people's access to routine healthcare including weekly haemodialysis schedules for chronic kidney patients and routine check-ups of cardiovascular patients [21]. It is well known that monitoring of patients on antimalarials is important even for those deemed at low risk. Albeit low-risk, electrocardiographic abnormalities may be serious, particularly in people who have serum electrolyte abnormalities, or are on other medications, including diuretics, macrolides, other QT-prolonging drugs and those who have chronic renal insufficiency and heart diseases [19,20]. Even simple and widely prevalent conditions in slums like gastroenteritis may put patients at risk by causing dyselectrolytemia. Hence, we suggest that in case the implementation of long term CQ and HCQ mass prophylaxis in the community is imminent, it should only be given after people have had ECGs and a baseline QT interval measurement made with the determination of serum electrolyte levels. While a conventional 12 lead ECG would be ideal, there are approved, low cost, handheld, portable, pocket sized ECG devices available, which can be connected to a smartphone wirelessly via blue-tooth technology and used in this set up [22].

The intention of the Indian Council of Medical Research and the Brihanmumbai Municipal Corporation to protect the people from the pandemic through empiric and pre-emptive mass administration of CQ and HCQ to the community at large is fully acknowledged. However, transparency is vital, and trust must be taken seriously. Community engagement and understanding of what is being proposed is of paramount importance. A police-led forceful implementation of any public-health measures including rolling out of mass prophylaxis of CQ and HCQ in the present situation to the urban slum communities is unwise to say the least.

Public distrust is already high after a series of unfortunate incidents provoked India's national cabinet to revise a part of The Epidemic Diseases Act of 1897, in a meeting held on 22nd April, to include harsh punishments against violent offenders who damage and attack the healthcare service personnel during the times of epidemic by booking them for a cognizable and non-bailable offence with hefty penalties and seven years of imprisonment [23].

From the people's perspective, a humanitarian community intervention plan is needed for the urban slums, replacing the spatial sealing off plans and preceding the rolling out of CQ and HCQ mass prophylaxis.

In our suggestions:

- Environmental and sewage surveillance should be employed in predicting the presence as well as the burden of COVID-19 [24], for identifying the urban slums at maximum risk.
- Self-collected "gargle and spit" method samples for adults as well as children above 5 years are comparable or even more sensitive than cough inducing and uncomfortable throat and nasal swab samples for respiratory viral pathogens [25–27]. If employed, with contactless transfer into triple-layered, transparent zipper bags, this method shall also be much less bio-hazardous during the sample collection in the slums by community healthcare workers without the need for repeated changing of personal protective equipment.
- Significantly enhanced numbers of testing among the slum dwellers is to be coupled with early isolation of the infected cases to a common community quarantine facility, which is secure, transparent in terms of governance and internal affairs, and dignified for the quarantined individuals with necessary wage compensation for the quarantined individuals, majority of whom are informal and unorganized sector workers.
- Early identification of infected high-risk individuals in the slums and community quarantine centres, with aggressive triaging and escorted referrals of such cases to secondary or tertiary care centres will further help in reducing the preventable mortalities.

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**Jim Thompson** • 2 years ago

And yet...India is doing spectacularly well given their population of 1.3 Billion. New York, where Governor Cuomo essentially blocked early HCQ use by issuing an executive order 23 March, now has 1,400 deaths/Million. India, where the ICMR promoted early use of HCQ for prophylaxis, has 2 deaths/Million. In NY, Cuomo's exec order forbade pharmacists from dispensing HCQ unless a patient was enrolled in a formal study. So if you got symptoms, first you had to get a PCR. Then wait 5-10 days for a result. Then persuade your practitioner to enroll you in a study. Result: no one in NY legally gets HCQ. They have the worst results in the world for number of patients/million who get COVID, and horrible case fatality rates (3X India's CFR). India (and Bangladesh), with their slums that are way more crowded than NY, cannot socially distance. Their elderly live with family and cannot isolate. Yet India is doing spectacularly well so far, hanging on by its fingernails to a death rate 1/700th that of NY. The answer may not be HCQ, but to ignore results while characterizing their HCQ approach as "baffling" is kind of absurd. Perhaps their HCQ approach is driven by ACTUAL OUTCOMES to DATE??!!

3 ^ | ▾ • Reply • Share ›

**Bob Enyart** → Jim Thompson • 2 years ago

Jim, you wrote that 3 months ago. I double checked and just confirmed the numbers for myself. Governor Cuomo all but banned HCQ and New York has seen 1,700 COVID deaths per million. India, with its population of 1.3 billion and densities for a large percent of the population far in excess of any NY overcrowding, has widely dispensed hydroxychloroquine and has seen 29 deaths per million. (To aet the cost in lives of NY's

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